Campus shuttle options. Audience question: Which shuttle option meets the minimum range at the lowest upfront cost?

Alder has the lowest upfront cost among the options that meet the 200 km minimum

Four shuttle options compared on the same criteria, in source order. Range in kilometres; costs in thousands of US dollars.

Criterion	Alder Lowest upfront cost meeting minimum	Birch	Cedar	Dogwood
Range (km)	210	240	190	220
Meets 200 km minimum?	**Yes**	**Yes**	**No**	**Yes**
Range difference from 200 km minimum (km, derived)	+10	+40	−10	+20
Upfront cost (USD thousands)	38	44	35	47
Estimated annual energy cost (USD thousands per year)*	4.2	3.9	4.5	3.8

* Assumes 18,000 km of service per vehicle per year. Excludes maintenance and charging-infrastructure costs.

> **Answer:** Alder (210 km, upfront cost 38 thousand USD) has the lowest upfront cost of the three options that meet the 200 km minimum. Birch costs 44 and Dogwood costs 47.
>
> **Not the best on every criterion:** Alder's estimated annual energy cost (4.2) is higher than Birch (3.9) and Dogwood (3.8). Birch has the longest range (240 km) but the second-lowest upfront cost among qualifying options at 44.
>
> **Cedar** has the lowest upfront cost (35) but its range of 190 km is 10 km below the minimum.

Caveats and assumptions

- This is a fictional scenario; all figures are invented for the evaluation.
- Costs are fictional planning estimates in thousands of US dollars. No uncertainty ranges were provided for any figure.
- The annual energy estimates assume 18,000 km of service per vehicle and exclude maintenance and charging-infrastructure costs.
- The answer uses upfront cost only. Annual energy cost is shown for context and is not part of the upfront-cost ranking.

Source: the procurement worksheet supplied with the brief. The procurement minimum is 200 km of range.